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EE 571

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HW1 Two-Port On-Chip Transformer Simulation

Introduction

This homework is designed to help students have hands-on experience in designing and electromagnetically simulating a two-port on-chip transformer using Cadence EMX and the TSMC 65 nm PDK. In this homework, students draw the layout of the transformer and simulate it to get the S-parameters of their transformer. Inductance, mutual inductance, coupling factor, quality factor, and self-resonant frequency are calculated from the S-parameters from the EMX simulation, and the EMX model is used in the Cadence Schematic testbench.

Transformer Layout

The design layout requirements for this homework are to design a 1:1 two-winding RF transformer using the TSMC 65nm PDK. The metal width should be 1 μm , and the minimum spacing between two metals should be 0.5 μm . I set the metal width to 1 μm as required and the minimum spacing to 0.7 μm due to the via size. Figure 1 shows the transformer with the port names and spacing.

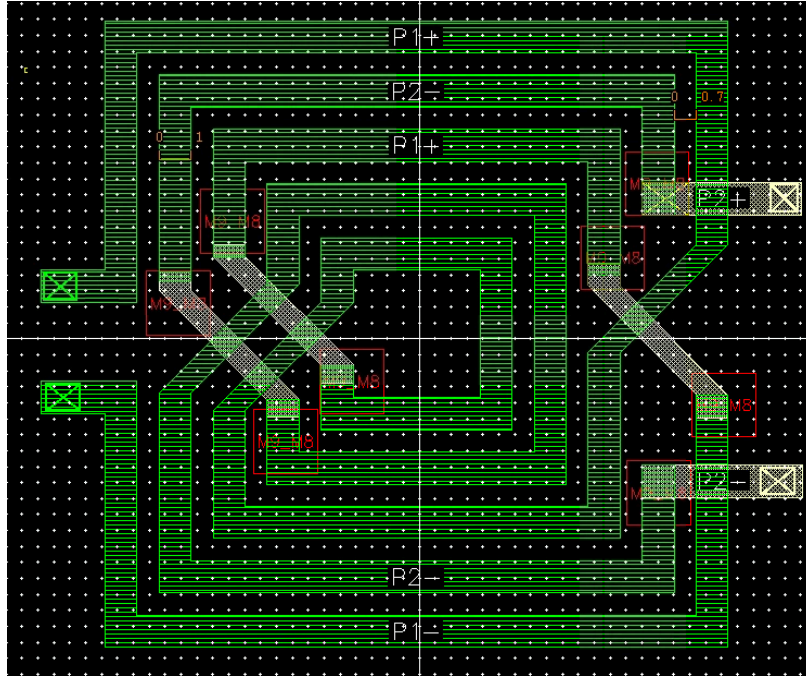


Figure 1. Screenshot of Transformer Layout

EMX Simulation

To run the EMX simulation, the signals are set first. P1+ is set to the signal, and P1- is set to its reference. Set P2+ to the other signal and P2- to its reference. The frequency sweep is set to 100 MHz to 30 GHz for the simulation. S-parameter data is saved after the simulation. Figure 2 shows plots of S_{11} , S_{12} , S_{21} , S_{22} . Figure 3 shows S_{11} and S_{22} only, and Figure 4 shows S_{12} and S_{21} only for better clarification. S_{12} and S_{21} are the same at every frequency. Figure 5 shows the phase of S_{21} .

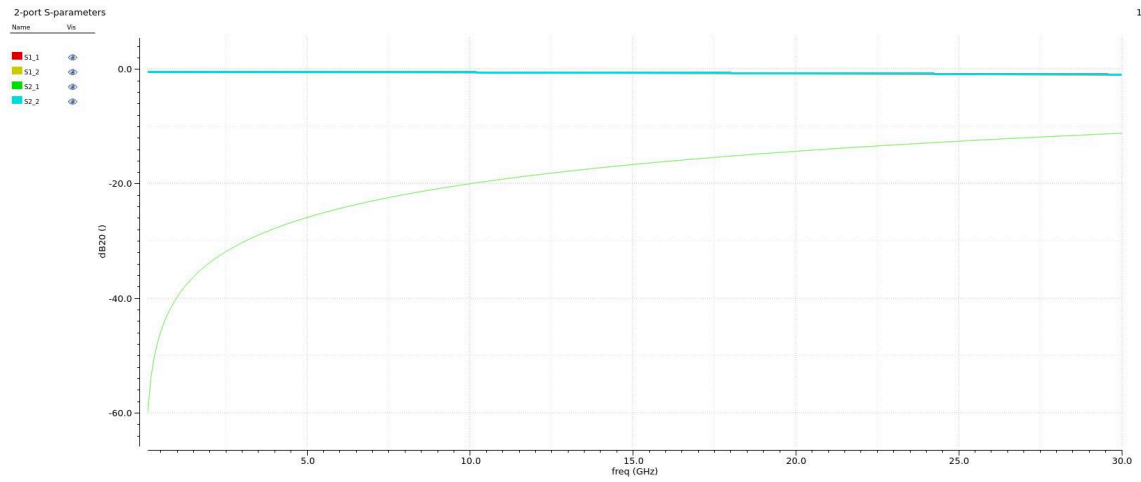


Figure 2. Plots of S_{11} , S_{12} , S_{21} , S_{22} .

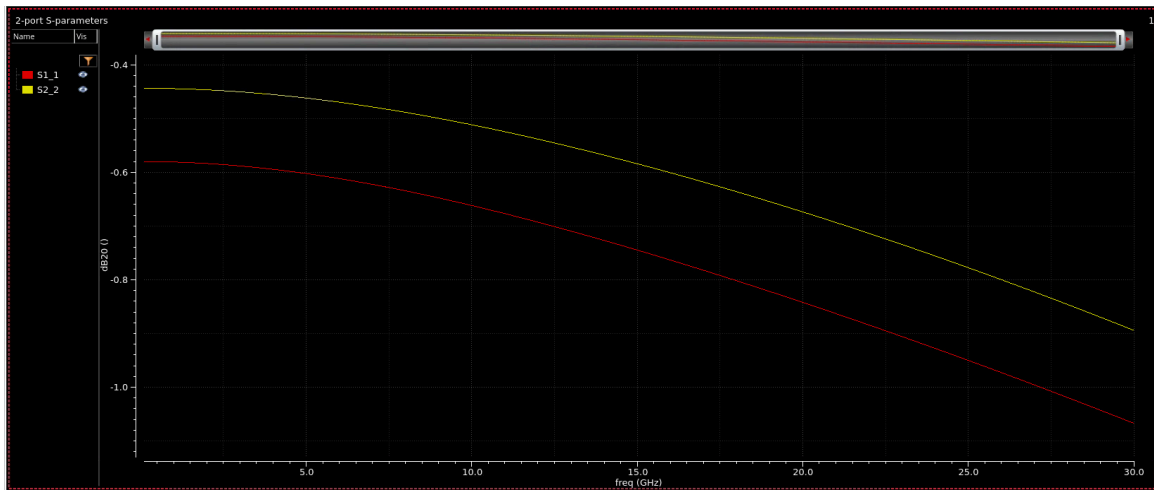


Figure 2. Plots of S_{11} and S_{22} .

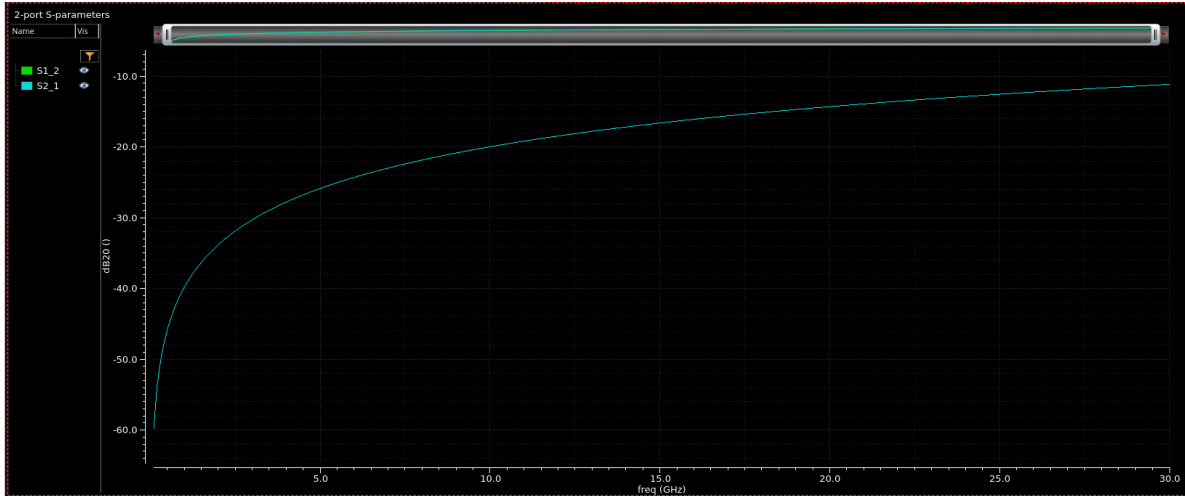


Figure 4. Plots of S_{12} and S_{21} .

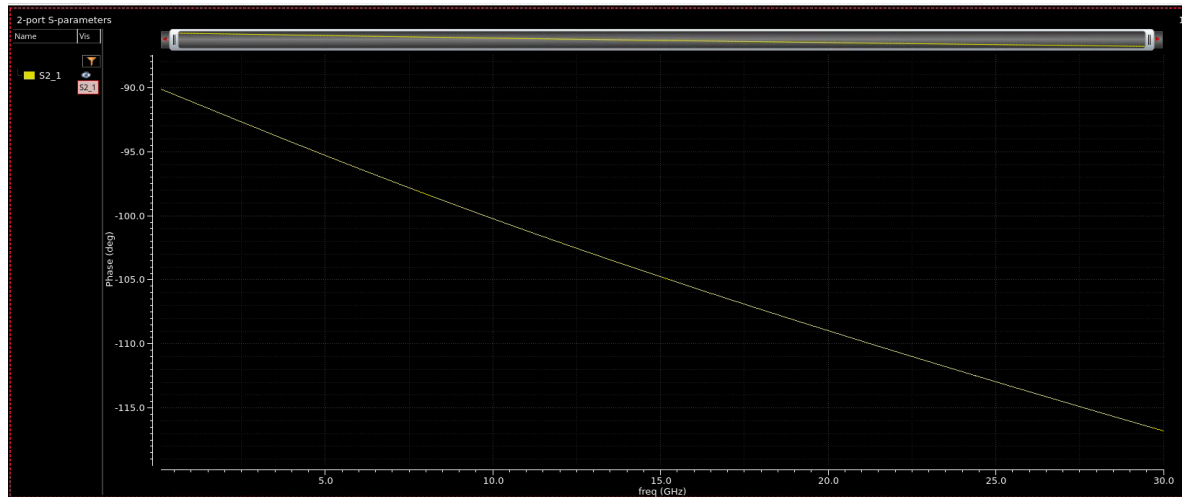


Figure 5. Plots of S_{21} Phase.

Calculating Inductance, Mutual Inductance, and Coupling Factor

S-parameter values are saved in a .s2p file after the simulation. Using this file and MATLAB, S-parameter values are converted to Z-parameters, and inductance, mutual inductance, and coupling factor are calculated. The MATLAB code is shown in Appendix A. At 5GHz, Inductance L1 is 0.0812 nH, L2 is 0.0462 nH, mutual inductance is -0.0431 nH, and the coupling

factor is 0.7037. Figure 6 shows plots of inductance, mutual inductance, coupling factor, and quality factor between 100 MHz and 30 GHz.

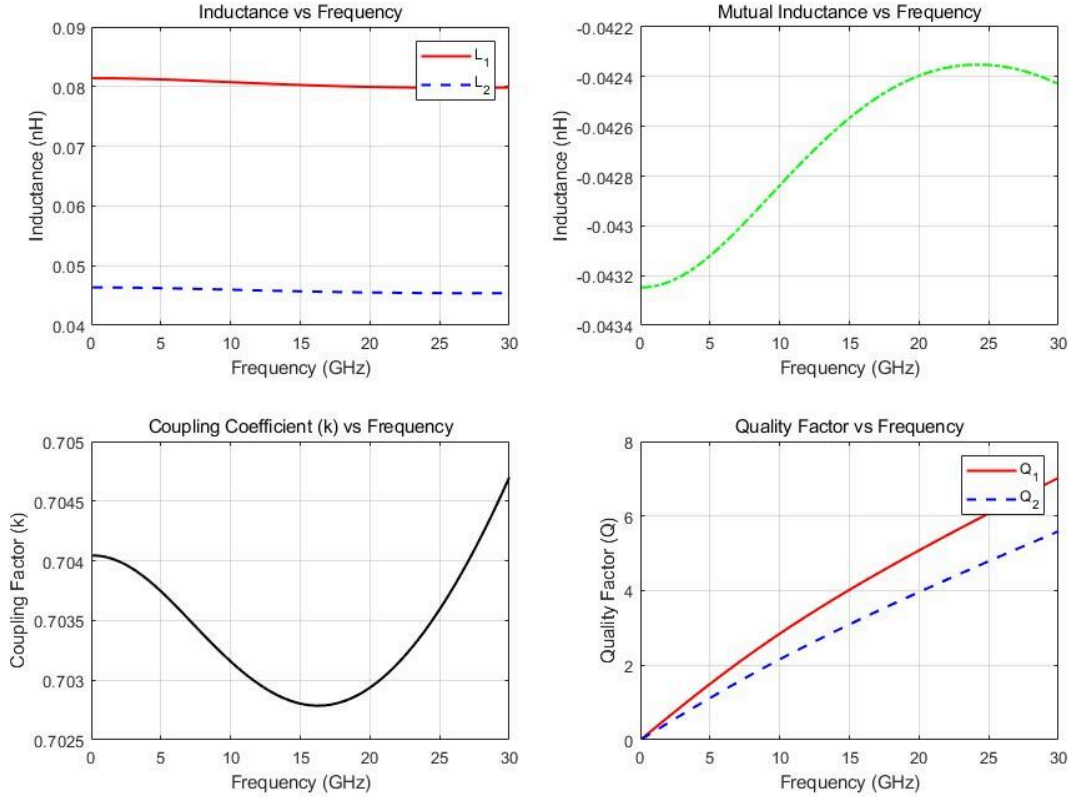


Figure 6. Plots of Inductance, Mutual Inductance, Coupling Factor, and Quality Factor

Self-Resonant Frequency

Self-resonant frequency, or SRF, can be found using the equation: $Im\{Z_{11}\} = 0$ for primary and $Im\{Z_{22}\} = 0$ for secondary. For MATLAB, the find function is used to find the frequency when the imaginary value of Z_{11} or Z_{22} is zero. However, the result of the MATLAB code says that SRF for both primary and secondary are located above 30 GHz. You can also look at the quality factor plots in Figure 6; Q is growing throughout the plots and does not decrease. This plot also shows that the SRF is located above 30 GHz, which is above our max limit of the simulation.

Cadence Schematic

To implement the EMX model, n2port in analogLib is used. In n2port's properties, the .s2p file is added as the S-parameter data file. Then, ports are added as in Figure 7, and the simulation is run.

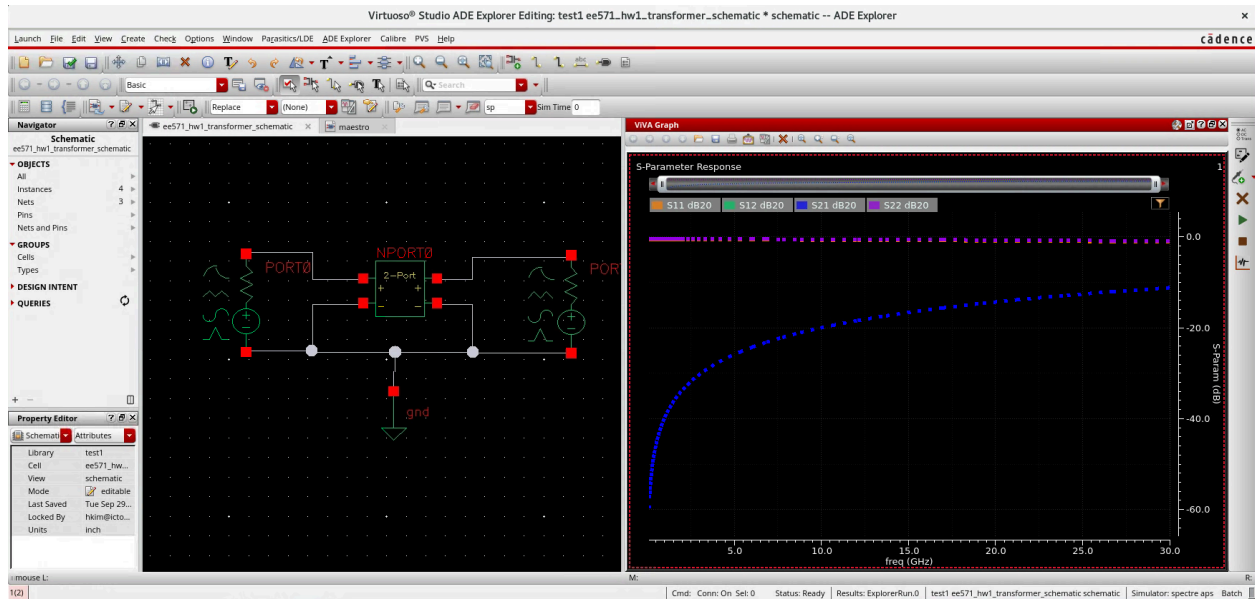


Figure 7. Schematic of EMX-Model and Result of Simulation

To check if the result of the simulation matches the original EMX file, three points of frequency - 5, 10, and 15 GHz - are selected. Figure 8 shows the selected points in the plots. The original values are extracted using MATLAB. The values of S_{11} in the simulation are -0.6 dB at 5 GHz, -0.663 dB at 10 GHz, and -0.747 dB at 15 GHz, where the original values are -0.58 dB at 5 GHz, -0.602 dB at 10 GHz, and -0.662 dB at 15 GHz. The values of S_{12} in the simulation are -26.141 dB at 5 GHz, -19.979 dB at 10 GHz, and -16.548 dB at 15 GHz, where the original values are -39.782 dB at 5 GHz, -25.854 dB at 10 GHz, and -19.968 dB at 15 GHz.



Figure 8. Plots of S-parameters with the Markers

Conclusion

The coupling factor is about 0.704 at 5 GHz and decreases at 15 GHz and increases again as frequency goes to 30 GHz. The average coupling factor is above 0.7; thus, I believe it has less loss. Also, as it goes to higher frequency, the coupling factor will increase, which makes a better transformation from primary to secondary. The quality factor also shows that this transformer performs better when the frequency increases. Since the self-resonant frequency is located above 30 GHz, this transformer is good to use for RF circuits or circuits that perform at higher frequencies.

APPENDIX A

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% Import Simulation Results
filename = 'ee571_hw1_transformer.s2p';
S_data = sparameters(filename);

freq = S_data.Frequencies;
freq_GHz = freq / 1e9;
Z_data = s2z(S_data.Parameters, S_data.Impedance);

% Convert to Z Parameter
Z11 = squeeze(Z_data(1,1,:));
Z12 = squeeze(Z_data(1,2,:));
Z21 = squeeze(Z_data(2,1,:));
Z22 = squeeze(Z_data(2,2,:));

% Calculating L1, L2, Mutual Inductance, and Coupling Factor
w = 2 * pi * freq;

L1 = imag(Z11) ./ w;
L2 = imag(Z22) ./ w;

M = imag(Z12) ./ w;
k = abs(M) ./ sqrt(L1 .* L2); % Since M was negative value due to dot convention, solved for |k|

Q1 = imag(Z11) ./ real(Z11);
Q2 = imag(Z22) ./ real(Z22);

% Plot
figure('Name', 'Transformer Parameters', 'Position', [100 100 1000 700]);

% (1) Inductance (L1, L2)
subplot(2,2,1);
plot(freq_GHz, L1*1e9, 'r-', 'LineWidth', 1.5); hold on;
plot(freq_GHz, L2*1e9, 'b--', 'LineWidth', 1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Inductance (nH)');
title('Inductance vs Frequency');
legend('L_1', 'L_2');

% (2) Mutual Inductance (M)
subplot(2,2,2);
plot(freq_GHz, M*1e9, 'g-', 'LineWidth', 1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Inductance (nH)');
title('Mutual Inductance vs Frequency');

% (3) Coupling Coefficient (k)
subplot(2,2,3);
plot(freq_GHz, k, 'k-', 'LineWidth', 1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Coupling Factor (k)');
title('Coupling Coefficient (k) vs Frequency');

% (4) Quality Factor (Q1, Q2)
subplot(2,2,4);
plot(freq_GHz, Q1, 'r-', 'LineWidth', 1.5); hold on;
plot(freq_GHz, Q2, 'b--', 'LineWidth', 1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Quality Factor (Q)');
title('Quality Factor vs Frequency');
legend('Q_1', 'Q_2');

% Calculating SRF
srf_idx1 = find(imag(Z11) <= 0, 1, 'first');
srf_idx2 = find(imag(Z22) <= 0, 1, 'first');

if ~isempty(srf_idx1)
    f_SRF1 = freq_GHz(srf_idx1);

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fprintf('Primary SRF (f_SRF1) : %.2f GHz\n', f_SRF1);
else
    fprintf('Primary SRF (f_SRF1) : Over 30 GHz (Not in the Simulation)\n');
end

if ~isempty(srf_idx2)
    f_SRF2 = freq_GHz(srf_idx2);
    fprintf('Secondary SRF (f_SRF2) : %.2f GHz\n', f_SRF2);
else
    fprintf('Secondary SRF (f_SRF2) : Over 30 GHz (Not in the Simulation)\n');
end

%-----RESULT-----
% Primary SRF (f_SRF1) : Over 30 GHz (Not in the Simulation)
% Secondary SRF (f_SRF2) : Over 30 GHz (Not in the Simulation)
% Matching the Schematic testbench
S = sparameters('ee571_hw1_transformer.s2p');
freq = S.Frequencies;
s_matrix = S.Parameters;
s11_dB = 20 * log10(abs(squeeze(s_matrix(1,1,:))));
s12_dB = 20 * log10(abs(squeeze(s_matrix(1,2,:))));
[~, idx_5G] = min(abs(freq - 5e9));
[~, idx_10G] = min(abs(freq - 10e9));
[~, idx_15G] = min(abs(freq - 15e9));
fprintf(' 5 GHz : S12 = %.4f dB, S11 = %.4f dB\n', s12_dB(idx_1G), s11_dB(idx_1G));
fprintf(' 10 GHz : S12 = %.4f dB, S11 = %.4f dB\n', s12_dB(idx_5G), s11_dB(idx_5G));
fprintf(' 15 GHz : S12 = %.4f dB, S11 = %.4f dB\n', s12_dB(idx_10G), s11_dB(idx_10G));
%-----RESULT-----
% 5 GHz : S12 = -39.7822 dB, S11 = -0.5809 dB
% 10 GHz : S12 = -25.8539 dB, S11 = -0.6023 dB
% 15 GHz : S12 = -19.9681 dB, S11 = -0.6618 dB

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